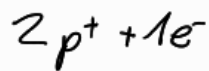
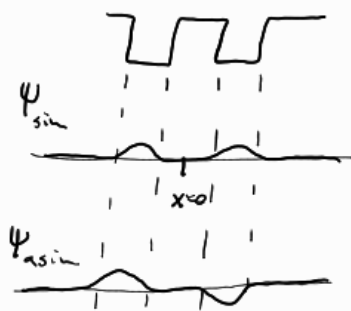


ion  $H_2^+$ : vezava s kvantnim tuneliranjem



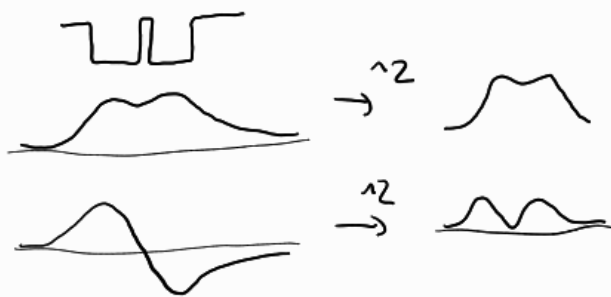
• toj model: 2 končni pot. jami



$\left\{ \begin{array}{l} \text{Ti 2 možnosti, ker } e^- \\ \text{z enako verjetnostjo} \\ \text{v levi ali desni jami.} \end{array} \right.$

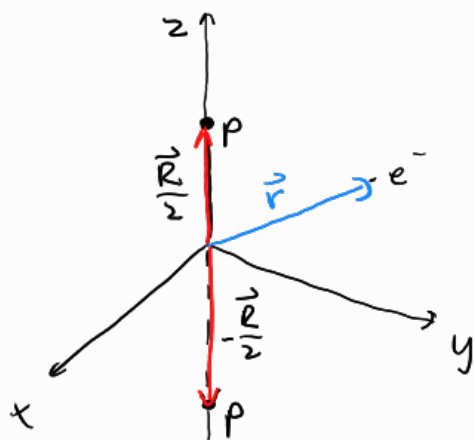
$\Rightarrow |\Psi|^2$  simetrično okrog  $x=0$

• jami bližje skupaj

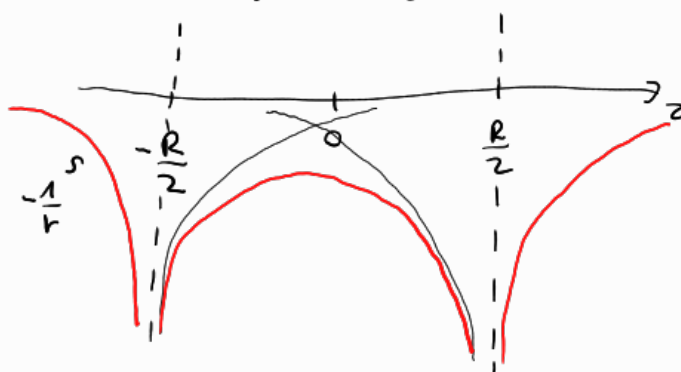


$\left\{ \begin{array}{l} \text{simetrična VF ima med jedroma} \\ \text{večjo verj. gostoto} \end{array} \right.$

• uporabimo Born-Oppenheimerjev približek: protona sta pri miru,  $e^-$  se giblje



Potencial vzdolž osi  $z$ :



SSE za  $e^-$ :

$$\left( -\frac{\hbar^2}{2m_e} \nabla_r^2 - \frac{ke_o^2}{|\vec{r} - \frac{\vec{R}}{2}|} - \frac{ke_o^2}{|\vec{r} + \frac{\vec{R}}{2}|} + \frac{ke_o^2}{R} \right) \Psi(\vec{r}) = E \Psi(\vec{r})$$

$\uparrow$   
 odbojna  
 sila med  
 protonoma  
 (neodl. od  $\vec{r}$ )  
 $\rightarrow$  zamik/offset

Kako konstruirati VF  $\Psi(\vec{r})$ ?

$$P(\vec{r}) = P(-\vec{r}) \rightarrow |\Psi(\vec{r})|^2 = |\Psi(-\vec{r})|^2 \Rightarrow \Psi(-\vec{r}) = \pm \Psi(\vec{r})$$

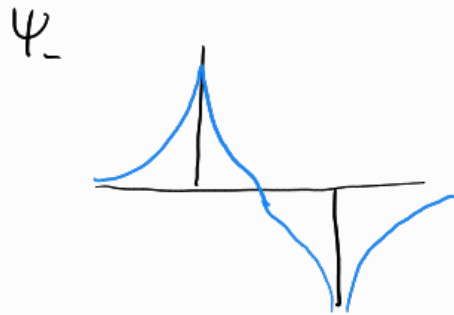
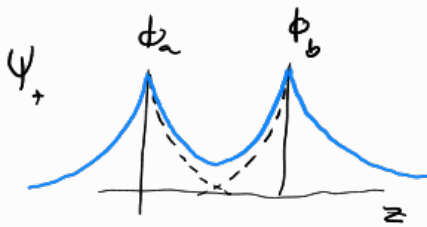
Imamo 2 možnosti:

$$\Psi_+(\vec{r}) = \frac{1}{\sqrt{2}} \left( \phi_{1s}(\vec{r} - \frac{\vec{R}}{2}) + \phi_{1s}(\vec{r} + \frac{\vec{R}}{2}) \right) = \frac{1}{\sqrt{2}} (\phi_a + \phi_b)$$

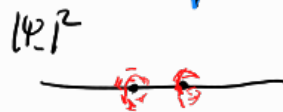
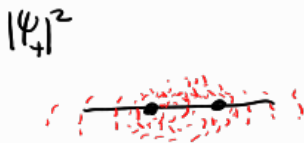
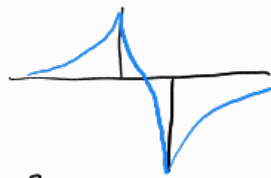
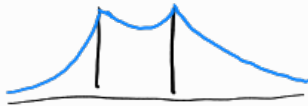
(približek) VF za osn. stanje H-atomov

$$\Psi_-(\vec{r}) = \frac{1}{\sqrt{2}} (\phi_a - \phi_b)$$

$$\phi_{1s}(r) \propto e^{-r/a_0}$$



$\phi_a, \phi_b$  še bližje



Napoved: NI STABILNO

$$E_{\pm} = \int \Psi_{\pm}^* \left[ \underbrace{-\frac{\hbar^2}{2m_e} \nabla^2}_{T} + V(\vec{r}) \right] \Psi_{\pm}(\vec{r}) d^3\vec{r}$$

$$T, V_a = -\frac{ke_0^2}{|\vec{r} - \vec{R}_a|}, V_b = -\frac{ke_0^2}{|\vec{r} - \vec{R}_b|}, V_R = -\frac{ke_0^2}{R} \rightarrow V = V_a + V_b + V_R$$

$$\begin{aligned} E_{\pm} &= \frac{1}{2} \int (\phi_a \pm \phi_b) (T + V_a + V_b + V_R) (\phi_a \pm \phi_b) d^3\vec{r} = \\ &= \frac{1}{2} \int (\phi_a \pm \phi_b) \left[ \underbrace{(T + V_a)\phi_a}_{E_{1s}\phi_a} \pm \underbrace{(T + V_b)\phi_b}_{E_{1s}\phi_b} \right] d^3\vec{r} + \\ &+ \frac{1}{2} \int \phi_a^2 (V_b + V_R) d^3\vec{r} + \frac{1}{2} \int \phi_b^2 (V_a + V_R) d^3\vec{r} + \\ &\pm \frac{1}{2} \int \phi_a (V_a + V_R) \phi_b d^3\vec{r} \pm \frac{1}{2} \int \phi_b (V_b + V_R) \phi_a d^3\vec{r} \end{aligned}$$

$$(T + V)\phi = E\phi$$

$$G \equiv \int \phi_a^2 (V_b + V_R) d^3\vec{r} = \int \phi_b^2 (V_a + V_R) d^3\vec{r}$$

$$S \equiv \int \phi_a (V_a + V_R) \phi_b d^3\vec{r} = \int \phi_b (V_b + V_R) \phi_a d^3\vec{r}$$

↑  
prekrivalni integral

$$\rightarrow E_{\pm} = E_{1s} + G \pm S$$

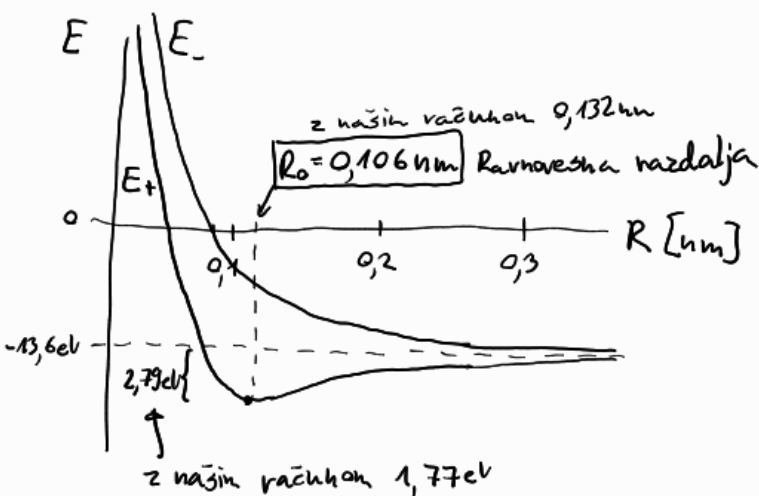
kot  $1e^-$  v  
polju enega  $p^+$

izkaže se:  $G(R \rightarrow \infty) \rightarrow 0$

$$G > 0$$

$$S(R \rightarrow \infty) \rightarrow 0$$

$$S < 0$$



• stanje  $E_-$  ima min v  $\infty \Rightarrow$  nevezano stanje (ni stabilno)

• stanje  $E_+$  ima min v  $R_0 \Rightarrow$  vezano stanje, vezano za 2,79 eV (če želimo razcepiti na H in  $p^+$ , moramo dovesti 2,79 eV energije)

• ker je  $E_+$  vezano  $\Rightarrow H^{2+} \exists$

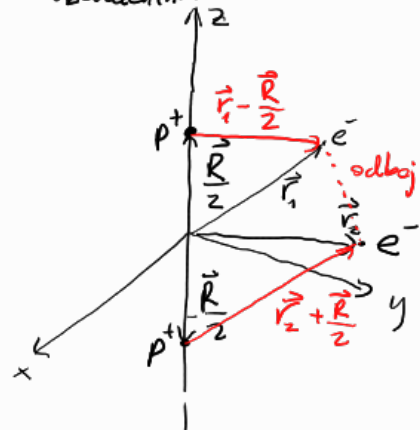
•  $\psi_+$  doda  $e^-$  gostoto med protona  $\rightarrow$  senčenje odboja

- problem obravnave z običajnimi vodikovimi VF:

pri  $R \rightarrow 0$  bi VF morala preiti v  $e^{-2r/r_B}$ , dobili pa smo  $2e^{-r/r_B}$  ker nastane He jedro

## Molekula $H_2$

• kovalentna vez



• potrebujemo dvoelektrno VF, gradimo postopoma:

1. približek: jedro skupaj ( $\vec{R} = 0$ ), antisimetrično (zagotovi  $\psi(2,1) = -\psi(1,2)$ )

$$\psi(1,2) = \underbrace{\phi_{1s}(\vec{r}_1) \phi_{1s}(\vec{r}_2)}_{\text{krajni del VF}} \underbrace{\chi^A(1,2)}_{\text{spinska VF}}$$

$$\chi^A(1,2) = \frac{1}{\sqrt{2}} (\uparrow_1 \downarrow_2 - \downarrow_1 \uparrow_2)$$

2. približek: protona razmaknemo

$$\psi(1,2) = \phi_{1s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 + \frac{\vec{R}}{2}) \chi^A(1,2)$$

• pomanjkljivosti:

1) ni antisimetrije ( $\psi(2,1) \neq -\psi(1,2)$ )

2) vsak  $p^+$  ima svoj "zasebni"  $e^-$ , nimamo možnosti, da bi bila oba  $e^-$  hkrati ob enem  $p^+$

Heitler-London

$$\rightarrow \psi_s^H(1,2) = \frac{1}{\sqrt{2}} \left( \phi_{1s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 + \frac{\vec{R}}{2}) + \phi_{1s}(\vec{r}_1 + \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 - \frac{\vec{R}}{2}) \right) \chi^s(1,2)$$

↑  
singletno spinsko stanje

• še vedno hiba: ne moremo imeti dveh  $e^-$  okrog istega  $p^+$

Alternativa: 1s krajinski del, s spinski del.

$$\psi_T^H(1,2) = \frac{1}{\sqrt{2}} \left( \phi_{1s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 + \frac{\vec{R}}{2}) - \phi_{1s}(\vec{r}_1 + \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 - \frac{\vec{R}}{2}) \right) \underbrace{\chi_{m_s}^s(1,2)}_{\text{simetrično na } 1 \leftrightarrow 2}$$

↑  
tripletno spinsko stanje

$$\chi_{m_s}^s = \begin{cases} \uparrow_1 \uparrow_2; m_s = 1 \\ \frac{1}{\sqrt{2}} (\uparrow_1 \downarrow_2 + \downarrow_1 \uparrow_2); m_s = 0 \\ \downarrow_1 \downarrow_2; m_s = -1 \end{cases}$$

→ Katero stanje da vezavo in katero ne?

$$P_{S,T}(1,2) = \frac{1}{2} \left[ \phi_{1s}^2(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}^2(\vec{r}_2 + \frac{\vec{R}}{2}) + \phi_{1s}^2(\vec{r}_2 - \frac{\vec{R}}{2}) \phi_{1s}^2(\vec{r}_1 + \frac{\vec{R}}{2}) \right. \\ \left. + 2 \left( \phi_{1s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 + \frac{\vec{R}}{2}) \right) \left( \phi_{1s}(\vec{r}_2 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_1 + \frac{\vec{R}}{2}) \right) \right]$$

←  $\psi_s^H$   
←  $\psi_T^H$

• slutimo: + pri S bo okrepil vezavo:  $e^-$  v prostoru med jedroma se sicer odbijata, toda močnejši vpliv je privlak protonov.

• izračunamo pričakovano vr. E v stanjih  $\psi_s^H, \psi_T^H$ :

$$E_{S,T} = 2E_{1s} + G' \pm S'$$

prekrivalni (oz. izmenjalni) del

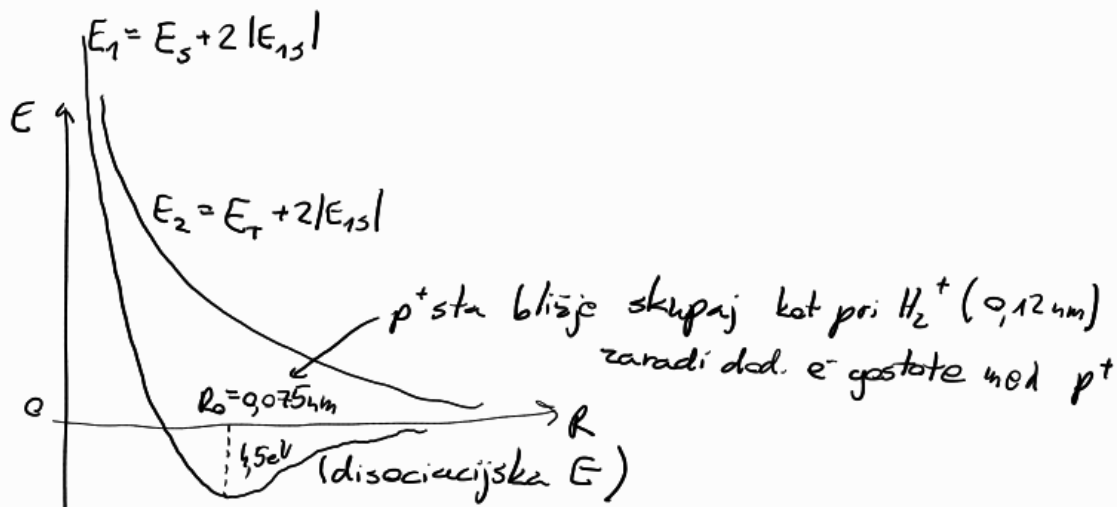
S' vsebuje člene oblike

$$\iint \phi_{1s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 + \frac{\vec{R}}{2}) \underbrace{V(\vec{r}_1, \vec{r}_2)}_{\text{vsebuje vse pot. člene, ki se pojavijo v energ. operatorju}} \phi_{1s}(\vec{r}_2 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_1 + \frac{\vec{R}}{2}) d^3\vec{r}_1 d^3\vec{r}_2$$

vsebuje vse pot. člene, ki se pojavijo v energ. operatorju

Ugotovitev:  $E_S < E_T$ ,  $E_S(R)$  ima minimum kot funkc. R,  $E_T(R)$  pa nima (oz. ima v  $R \rightarrow \infty$ )

⇒  $E_S$  je vezano stanje,  $E_T$  pa ne.



→ pri  $H_2^+$  je bila 2,7 eV. Če se  $e^-$  nebi odbijala med sabo, bi prihranili  $2 \times 2,7 \text{ eV} = 5,4 \text{ eV}$ .

3. približek oz. izboljšava: možnost, da oba  $e^-$  hkrati na enem  $p^+$  (neke vrste ionska slika  $H^- - H^+$  ali  $H^+ - H^-$ )

$$\psi_3(1,2) = \psi_s^{H_2}(1,2) + \gamma \frac{1}{\sqrt{2}} \left[ \phi_{1s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 - \frac{\vec{R}}{2}) + \phi_{1s}(\vec{r}_1 + \frac{\vec{R}}{2}) \phi_{1s}(\vec{r}_2 + \frac{\vec{R}}{2}) \right] \chi^A(1,2)$$

• izkaže se:  $\gamma = 0,2 \Rightarrow$  ionska komp. vezi v  $H_2$  ima nekaj % vloge,  $\psi_s^{H_2}$  je dominantna

### Opomba glede žlahtnih plinov

Ko skušamo spojiti žlahtni plin + nek atom (npr. H in He), nastopi Paulijeva prepoved: spin atoma se "tepe" s spinom enega v žlahtnem plinu

$\Rightarrow \nexists H-He, He-He, He-Ar$

• toda  $Ar_2 \exists$  (odgovorna Van der Waalsova interakcija)

• pri zelo nizkih T  $\exists$  He-He dimer,  $R_0 = 5,2 \text{ nm}$ ! (najv. znana 2-atomna molekula)

### Ionska vez

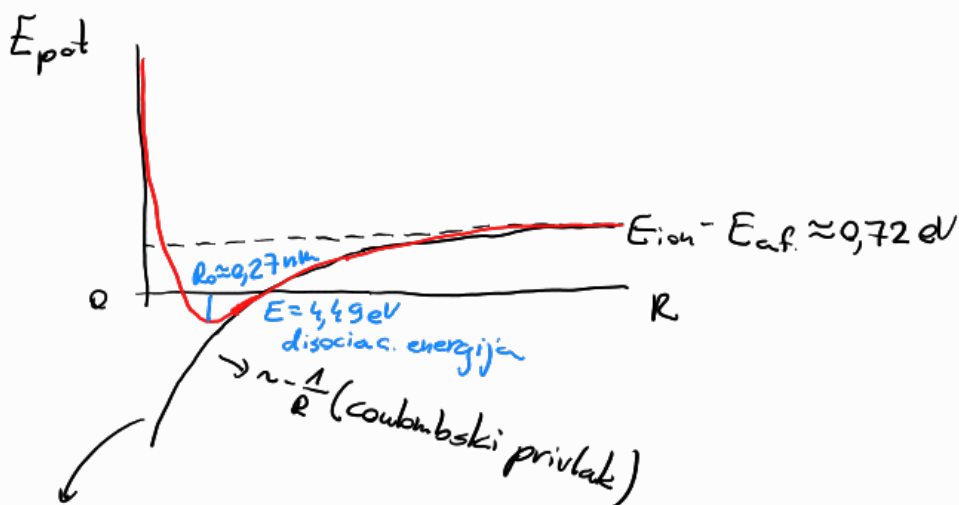
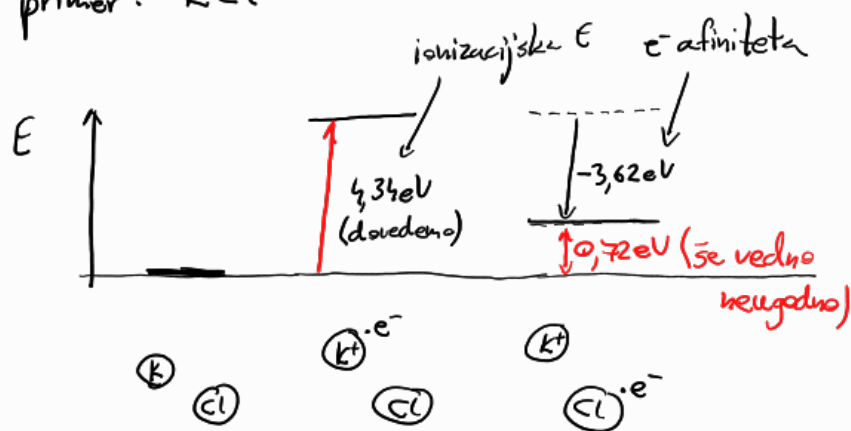
• tvorijo jo: atomi z  $1e^-$  na zadnji lupini (alkalni atomi)

in halogeni ( $1e^-$  manjka do polne zadnje lupine)

• alkalni atom odda  $1e^-$ , halogen ga sprejme

$\Rightarrow$  elektrostatiski privlak med  $+ve$  in  $-ve$  ionom

primer: KCl



- ko dajemo atoma vedno bolj skupaj:
  - nastane elektrostatični odboj med  $e^-$
  - Paulijeva prepoved (preveč  $e^-$  za pre malo stanj)

$$V(R) = -\frac{ke_0^2}{R} + V_{\text{odb}} + E_{\text{ioniz.}} - E_{\text{afin.}}$$

$\rightarrow$  običajno  $AR^{-n}$  ali  $Ae^{-BR}$   
 $n \gg 1$

Zgled: LiF

$$\left. \begin{array}{l} E_{\text{ion}} = 5,4 \text{ eV} \\ E_{\text{af}} = 3,4 \text{ eV} \end{array} \right\} \text{ strošek je } 2 \text{ eV, ampak ga kompenzira elektrost. privlak } \text{Li}^+ - \text{F}^-$$

el. dipolni moment?

idealno:  $e^-$  z Li ves čas na F

$$p_e = -e_0 \left(-\frac{R_0}{2}\right) + e_0 \frac{R_0}{2} = e_0 R_0$$

$$R_0 = 0,156 \text{ nm} \rightarrow p_e = 2,5 \cdot 10^{-29} \text{ Asm}$$

$$\text{iz eksp.: } p_e = 2,11 \cdot 10^{-29} \text{ Asm}$$

(0,85 pričakovane)

$\Rightarrow e^-$  ni samo na F

• v VF dopustimo tri možnosti: koval. + 2 ionski konf.

$$\Psi(1,2) = \left\{ A \frac{1}{\sqrt{2}} \left[ \phi_{2s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{2p}(\frac{\vec{r}_1 + \vec{R}}{2}) + \phi_{2s}(\frac{\vec{r}_1 - \vec{R}}{2}) \phi_{2p}(\vec{r}_1 + \frac{\vec{R}}{2}) \right] \right. \\ \left. (Li^+ F^- \rightarrow) + B \phi_{2s}(\vec{r}_1 - \frac{\vec{R}}{2}) \phi_{2s}(\frac{\vec{r}_1 + \vec{R}}{2}) \right. \\ \left. (Li^+ F^- \rightarrow) + C \phi_{2p}(\vec{r}_1 + \frac{\vec{R}}{2}) \phi_{2p}(\frac{\vec{r}_1 + \vec{R}}{2}) \right\} \chi^1(1,2)$$

• privzamemo  $B=0$  (zelo malo verj., da F odda  $e^-$  Li)

$$\rightarrow A^2 + B^2 + C^2 = 1$$

$$R_e = e_0 R_0 (C^2 - B^2) \rightarrow C = \sqrt{0.85} \approx 0.922$$

$$\rightarrow A = \sqrt{0.15} \approx 0.387 \Rightarrow \text{vez v LiF ima 15\% koval. značaj}$$

## Van der Waalsova interakcija

• omogoči vezavo žlehtnih plinov (atomi z zaklj. lupinami)

• osnovna: interakcija dveh el. dipolov



$$\langle \vec{p}_e \rangle = 0, \text{ toda } \sqrt{\langle \vec{p}_e^2 \rangle} \neq 0$$

• polje el. dipola:  $\vec{E}_d = \frac{\vec{p}_1}{4\pi\epsilon_0 R^3}$

$$\vec{p}_2 = \alpha \vec{E}_d$$

polarizirnost

$$\rightarrow V = -\vec{p}_2 \cdot \vec{E}_d =$$

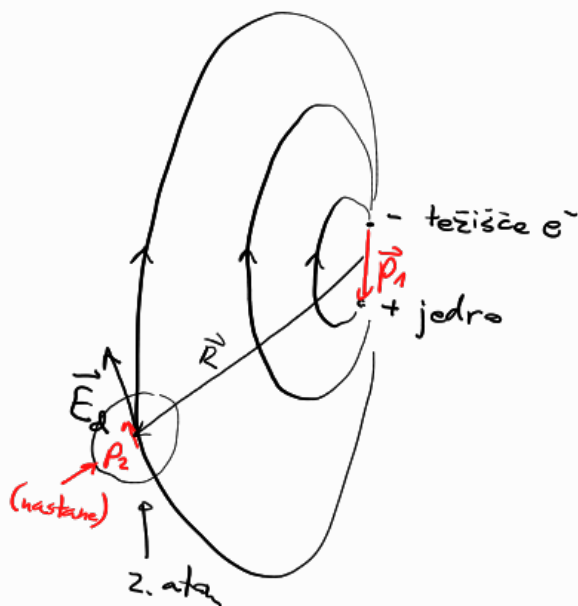
$$= -\alpha E_d^2 = -\alpha \frac{p_1^2}{(4\pi\epsilon_0)^2 R^6} \stackrel{p_1 = e_0 a}{=} -\alpha \frac{(e_0 a)^2}{(4\pi\epsilon_0)^2} \frac{1}{R^6}$$

• za občutek o  $\alpha$ :

$$\alpha = \frac{p}{E} \approx \frac{e_0 l}{\frac{1}{4\pi\epsilon_0} \frac{e_0}{l^2}} = 4\pi\epsilon_0 l^3 \approx 4\pi\epsilon_0 a^3 \text{ (polarizirnost } \propto \text{ prostornina sistema)}$$

$$\Rightarrow \boxed{F \propto -\frac{dV}{dR} \propto \frac{1}{R^7}}$$

$$\boxed{V \approx -\frac{e_0^2}{4\pi\epsilon_0 a} \left(\frac{a}{R}\right)^6} \text{ Van der Waalsova interakcija}$$





- VdW interakcija je šibka, toda ima bistveno daljši doseg kot (eksponentno padajoča) kovalentna vez

Dobro delujoča fenomenološka parametrizacija:

$$V_L(R) = 4\epsilon \left[ \left( \frac{\sigma}{R} \right)^{12} - \left( \frac{\sigma}{R} \right)^6 \right]$$

(Lennard-Jones)      ↓      ↓  
odboj      privlak

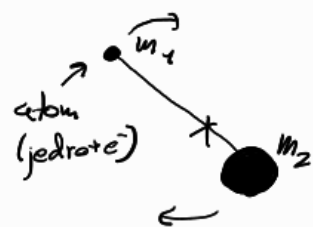
|    | $\sigma [\text{nm}]$ | $\epsilon [\text{eV}]$ |
|----|----------------------|------------------------|
| He | 0,256                | $8,79 \cdot 10^{-4}$   |
| Ne | 0,275                | $3,08 \cdot 10^{-3}$   |
| Ar | 0,340                | $1,05 \cdot 10^{-2}$   |
| Kr | 0,368                | $1,44 \cdot 10^{-2}$   |
| Xe | 0,407                | $1,94 \cdot 10^{-2}$   |

### Vzbujena stanja in spektri molekul

- rotacija in vibracija
- samo dvoatomne molekule

- vezavna šibka, raste z vel. atoma, ker se več polarizirnost

#### 1) Rotacija

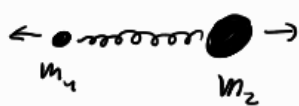


$$E_{\text{rot}} = \frac{\vec{L}^2}{2J} = \frac{\hbar^2 l(l+1)}{2\mu R_0^2}$$

$\mu = \frac{m_1 m_2}{m_1 + m_2}$   
 reducirana masa      ravnotežna razd. med jedroma

$$E_{\text{rot}} \approx 10^{-3} \text{ eV}$$

#### 2) Vibracija



⇒ 1D LHO, SSE:

$$-\frac{\hbar^2}{2\mu} \frac{1}{R^2} \frac{d}{dR} \left( R^2 \frac{d\psi}{dR} \right) + V(R) \psi(R) = E_{\text{vib}} \psi(R)$$

$$U(R) = R \psi(R)$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2 U}{dR^2} + V(R) U = E_{\text{vib}} U, \quad V(R) = \frac{1}{2} k R^2$$

boljši potencial (Morse):

$$V(R) = A \left[ e^{-2a(R-R_0)} - 2e^{-a(R-R_0)} \right] \quad (\text{tudi s tem je SSE analitično rešljiva})$$

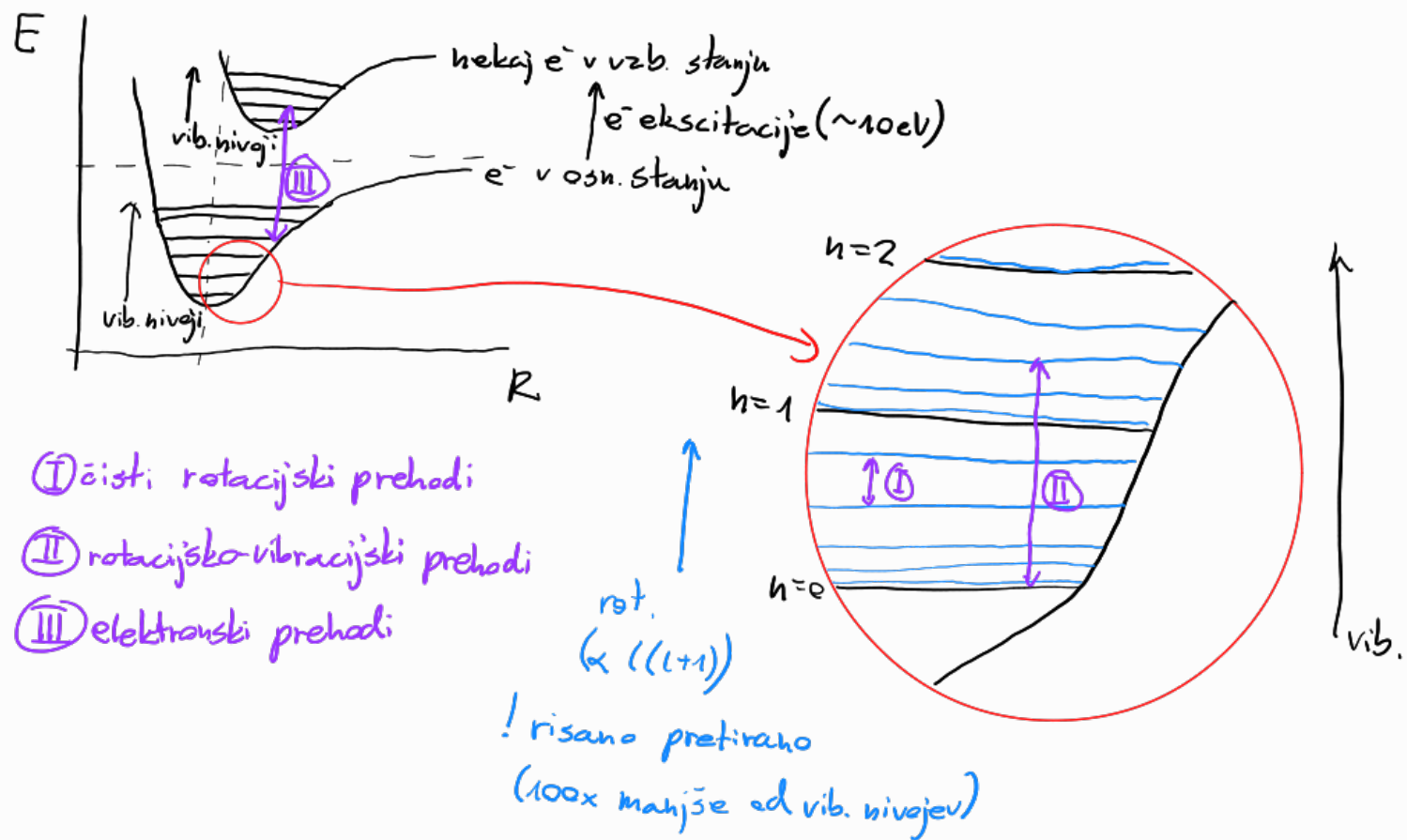
- $A, a, R_0$  odv. od molekule

$$\rightarrow V_k'' = k = V_n'' = 2a^2 A$$

$$\rightarrow E_{\text{vib}} \approx 10^{-1} \text{ eV}$$



# Spektri (dvoat. molekul)



$$E = E_{\text{el}} + h\nu_0\left(n + \frac{1}{2}\right) + B l(l+1)$$

$\downarrow$  el.       $\downarrow$  vib.       $\downarrow$  rot.

① absorpcija:  $l \rightarrow l+1$

$$h\nu \approx B[l(l+1) - l(l-1)] = \underline{2Bl}$$

emisija:  $l \rightarrow l-1$

$$h\nu \approx B[l(l+1) - (l-1)l] = \underline{2Bl}$$

→ spekter je ekvidistančen, v obeh primerih  $h\nu \propto l$

② najpogostejše:  $\Delta l = \pm 1$  (izb. pravilo za LHO)  
 $\Delta l = \pm 1$  (izb. pravilo za rotator)



absorpcija  $n=0 \rightarrow n=1$ ,  $l \rightarrow l+1$  (izberemo to komb.)

$$h\nu_R = h\nu_0\left(1 + \frac{1}{2}\right) + B(l+1)(l+2) - h\nu_0\left(0 + \frac{1}{2}\right) - B l(l+1) = h\nu_0 + 2B(l+1) \rightarrow \underline{\text{veja R}}$$

$n=0 \rightarrow n=1$ ,  $l \rightarrow l-1$

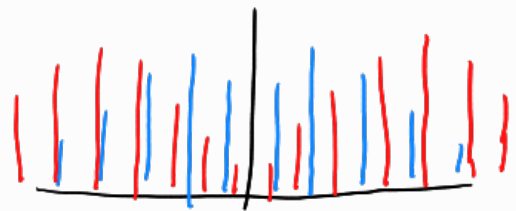
$$h\nu_P = h\nu_0\left(1 + \frac{1}{2}\right) + B(l-1)l - h\nu_0\left(\frac{1}{2}\right) - B l(l+1) = h\nu_0 - 2Bl \rightarrow \underline{\text{veja P}}$$

• iz  $\hbar\omega_0$  lahko določimo  $k$  ( $\omega_0 = \sqrt{\frac{k}{\mu}}$ )

$$n(\epsilon_l) = g(\epsilon_l) e^{-\epsilon_l/k_B T} \approx$$

$$= \underbrace{(2l+1)}_{\text{degeneracija}} \exp \left[ - \left( \underbrace{\frac{1}{2} \hbar \omega_0}_{\text{vib.}} + \underbrace{B l(l+1)}_{\text{rot.}} \right) / k_B T \right]$$

• pri  $T \ll$  so zasedena stanja  $z \ll$   
in obratno



nizke  $T$   
visoke  $T$

- III • spremeni se  $V(R) \rightarrow$  spremeni se  $R_0 \rightarrow$  spremeni se  $k$   
 $\Rightarrow$  spektri niso več ekvidistančni  
 • členi se ne odštejejo ( $\omega_0 \neq \omega'_0, B \neq B'$ )